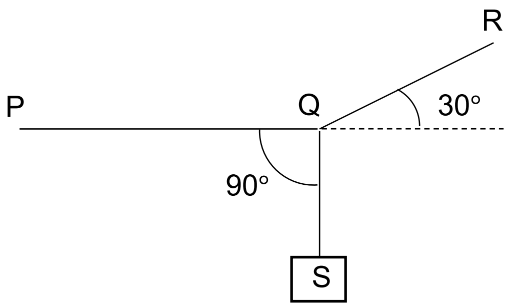
8	In the diagram below, a body S of weight *W* hangs vertically by a thread tied at Q to the string PQR.  If the system is in equilibrium, what is the tension in the section PQ?

A	$W\,cos30°$	**B**	$W\,cos60°$	**C**	$W\,tan30°$	**D**	$W\,tan60°$

Ans: D

[Turn over

Vertical component of tension = W
Hence tension in QR = $W/sin30°$
Tension in PQ = horizontal component of tension in QR = $\frac{W}{sin30°} \times cos30° = Wcot30° = Wtan60°$

9	Two farmers use an electrically powered elevator to lift bales of hay. All the bales of hay have